

Youngmin Kim

M.S. Student at KAIST

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Education

- Sep. 2026 – **M.S. in Artificial Intelligence**, KAIST, Seoul, Korea.
Present Kim Jaechul Graduate School of AI
Advisor: Jong Chul Ye
Concentration: 3D Vision and Robotics
- Feb. 2023 – **B.S. in Computer Science and Engineering**, Korea University, Seoul, Korea, GPA: 4.3 / 4.5.
Aug. 2026 Department of Computer Science and Engineering
Advisor: Jaehoon Lee
Early graduation (3.5 years)

Research Interests

3D Vision, Robotics, Vision-Language-Action (VLA), World Action Models (WAM), Diffusion Models, Generative Modeling, Medical Imaging

Publications

- [1] Sunwoo Park*, Wonbin Lee*, Seonghyun Jin*, **Youngmin Kim***, Jangho Park, Jong Chul Ye, "Dynamic Manipulation with World-Action Models via Counterfactual Planning", *arXiv* (2026), [arXiv:2609.33172](#), [project page](#)
(Under review)
- [2] Taesung Kwon, Jangho Park, Sunwoo Park, **Youngmin Kim**, Seonghyun Jin, Youngjun Jun, Kyumin Choi, Jong Chul Ye, "Don't Throw Away the Tail: Action Upcycling for Policy Acceleration", *arXiv* (2026), [arXiv:2609.34911](#)
(Under review)
- [3] Jeongsol Kim, Youngjun Jun, Kyumin Choi, **Youngmin Kim**, Seonghyun Jin, Sunwoo Park, Jangho Park, Kwanyoung Kim, Jong Chul Ye, "Adjoint Guidance Flow: Amortized Critic Guidance for VLA Policies", *arXiv* (2026), [arXiv:2609.34944](#)
(Under review)
- [4] Seonghyun Jin*, **Youngmin Kim***, Sunwoo Park*, Jong Chul Ye, "CRePE: Curved Ray Expectation Positional Encoding for Unified-Camera-Controlled Video Generation", *Advances in Neural Information Processing Systems (NeurIPS)* (2026), [arXiv:2605.12938](#)
(Poster; earlier version at the ECCV Workshop on 3D in the Era of World Models)
- [5] **Youngmin Kim***, Jaeyun Shin*, Jeongchan Kim*, Taehoon Lee, Jaemin Kim, Peter Hsu, Jelle Veraart, Jong Chul Ye, "DMD-augmented Unpaired Neural Schrödinger Bridge for Ultra-Low Field MRI Enhancement", *arXiv* (2026), [arXiv:2603.03769](#)
- [6] **Youngmin Kim***, Jeongchan Kim*, Taehoon Lee*, Jaeyun Shin*, Suhyeon Lee, Jong Chul Ye, "Ultra-Low-Field Brain MRI Enhancement using Resfusion and Residual Artifact Suppression Network", *MICCAI ULF-EnC Challenge Workshop* (2026)
- [7] **Youngmin Kim**, Hogeon Park, Heonchang Yu, "Performance Analysis of Kubernetes Traffic Scheduling Algorithms in Homogeneous and Heterogeneous Environments", *KCC* (2025)

- [8] **Youngmin Kim**, Wonyeong Jang, Taehee Jeong, “Improving Visual Question Answering via Prompt-Level Adaptation and Knowledge-Driven Fine-Tuning: Solution of Meta CRAG-MM Challenge 2025”, *Proceedings of the KDD Cup Workshop on CRAG-MM* (2025)

* These authors contributed equally.

Research Experience

- Sep. 2026 – **Graduate Researcher**, *BISPL (Bioluminescence Imaging, Signal Processing, & machine Learning Lab)*, KAIST, Korea.
Present Advisor: Jong Chul Ye
- Jun. 2025 – **Research Intern**, *BISPL (Bioluminescence Imaging, Signal Processing, & machine Learning Lab)*, KAIST, Korea.
Aug. 2026 Advisor: Jong Chul Ye
- Jun. 2024 – **Research Intern**, *Distributed and Cloud Computing Lab*, Korea University, Korea.
Jun. 2025 Network latency in Kubernetes.

Awards & Honors

- Jan. 2025 **Academic Excellence Award**, Korea University, Korea
- 2024 – 2025 **Semester Highest Honors** ($\times 3$), Korea University, Korea
Feb. 2025, Aug. 2024, Feb. 2024
- Aug. 2023 **Semester Honors**, Korea University, Korea

Projects

- Jul. 2026 – **DPP — Dynamic Manipulation with World-Action Models**, *BISPL*, Korea.
Present Counterfactual planning for moving targets; see Publications [1].
- Jul. 2026 – **Robot Training and Evaluation Interface**, *BISPL*, Korea.
Present
- Jun. 2026 – **CRoPE — Unified-Camera-Controlled Video Generation**, *BISPL*, Korea.
Aug. 2026 Curved Ray Expectation Positional Encoding; see Publications [4].
- Aug. 2025 – **Medical Foundational Model**, *BISPL*, Korea.
Mar. 2026
- Jun. 2025 – **ULF to High-field MRI Image Enhancement**, *BISPL*, Korea.
Present
- Nov. 2025 – **Short Movie Using 3DGS**, *Korea University*, Korea.
Dec. 2025
- Dec. 2024 – **Kaist Map**, *KAIST Madcamp*, Korea.
Present Backend developer of the Kaist Map app.
- Jul. 2024 – **Execution Time Analysis of Kubernetes Scheduler Algorithms in Homogeneous and Heterogeneous Environments**, *Distributed and Cloud Computing Lab*, Korea.
Mar. 2025

Technical Skills

- Languages Python, C++
- Frameworks PyTorch
- Simulation MuJoCo, Isaac Sim, LIBERO
- Models Vision-Language-Action (VLA), diffusion models, video generation models